MATH 350-2 Advanced Calculus

W.R. Casper

Department of Mathematics
California State University Fullerton

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Outline

1. Real Analysis Lecture 18
   - Extreme Value Theorem
   - Continuity and Algebra
   - Connectedness
   - Uniform continuity

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Extreme value theorem

Theorem

Suppose (S, d_S) is a metric space and $f : S \to \mathbb{R}$ is a function (to Euclidean space).
If f is continuous on a compact subset $A \subseteq S$, then there exists $a, b \in A$ such that

$$f(a) = \inf f(A), \quad \text{and} \quad f(b) = \sup f(A).$$

Extreme value theorem

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The infimum and supremum of a set are adherent points of that set.
Therefore, since $f(A)$ is closed $y, z \in f(A)$.

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The infimum and supremum of a set are adherent points of that set.
Therefore, since $f(A)$ is closed $y, z \in f(A)$.
This means that there exists $a, b \in A$ such that $y = f(a)$ and $z = f(b)$. □

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Warm-up

Prove that the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) = x + y$$

is continuous. (Everything Euclidean)

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Prove that the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) = xy$$

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Prove that the function $f : \mathbb{R} \times (\mathbb{R} \backslash \{0\}) \to \mathbb{R}$ defined by

$$f(x, y) = x/y$$

is continuous. (Everything Euclidean)

Algebra and continuity

Theorem

Let (S, d) be a metric space and $f : S \to \mathbb{R}$ and $g : S \to \mathbb{R}$ be functions which are continuous at $a \in S$.

- *$f + g$ is continuous at a*
- *fg is continuous at a*
- *f/g is continuous at a as long as $g(a) \neq 0$*

Polynomials and Rational Functions

- constant functions are continuous
- $f(x) = x$ is continuous
- monomials x^n are continuous
- polynomials $a_0 + a_1 x + \cdots + a_n x^n$ are continuous
- rational functions $p(x)/q(x)$ are continuous on their domains

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Characterization of intervals

Theorem

If $I \subseteq \mathbb{R}$ is an interval, then there does not exist two nonempty, disjoint open sets $U, V \subseteq \mathbb{R}$ (Euclidean space), with

$$I \subseteq U \cup V.$$

Connected spaces

Definition

A subset A of a metric space (S, d) is called **disconnected** if there exists U, V with $U \cap A$ and $V \cap A$ nonempty, U and V are disjoint and open, and $A \subseteq U \cup V$.

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We call S **connected** if it is not disconnected.

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Examples:

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- the real line with the discrete metric $(\mathbb{R}, d_{disc})$ is disconnected
- euclidean space $\mathbb{R}^n$ is connected
- an interval $I \subseteq \mathbb{R}$ with the Euclidean metric is connected

Challenge!

Problem

Prove that the set $\mathbb{Q} \subseteq \mathbb{R}$ is disconnected.

Real line is connected

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- $U \cup V = \mathbb{R}$

In particular, U is a nonempty, proper subset of $\mathbb{R}$ which is both open and closed. □

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By the Representation Theorem for Open Sets:

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$$U = (a_1, b_1) \cup (a_2, b_2) \cup \dots$$

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In particular, b_1 is an accumulation point of U but not an element of U.
This contradicts the fact that U is closed. □

Images of connected sets

Theorem

Let (S, d_S) and (T, d_T) be metric spaces and $f : S \to T$ be a function.
If f is continuous and S is connected, then $f(S)$ is connected.

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Suppose $f(S)$ is disconnected.
Then $f(S) = U \cup V$ for nonempty, disjoint, open sets $U, V \subseteq f(S)$.

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Let (S, d_S) and (T, d_T) be metric spaces and $f : S \to T$ be a function.
If f is continuous and S is connected, then $f(S)$ is connected.

Proof.

Suppose $f(S)$ is disconnected.
Then $f(S) = U \cup V$ for nonempty, disjoint, open sets $U, V \subseteq f(S)$.
Then $f^{-1}(U)$ and $f^{-1}(V)$ are non-empty, disjoint, open sets and

$$f^{-1}(U) \cup f^{-1}(V) = f^{-1}(U \cup V) = f^{-1}(f(S)) = S.$$

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This is a contadiction. □

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We can make any interval as a continuous image:

$$f : \mathbb{R} \to \mathbb{R}$$

$$f(x) = \frac{b+a}{2} + \frac{(b-a)x}{1+x^2}$$

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$$f(\mathbb{R}) = (a, b), \quad a < b$$



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We can make any interval as a continuous image:

$$f : \mathbb{R} \to \mathbb{R}$$

$$f(x) = a + \frac{2(b-a)x^2}{1+x^4}$$

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$$f(\mathbb{R}) = (b, a], \quad a > b$$

Intermediate value theorem

Theorem (Intermediate Value Theorem)

Suppose $f : [a, b] \to \mathbb{R}$ *is a continuous function. Then for any value* y *between* $f(a)$ *and* $f(b)$*, there exists* $c \in [a, b]$ *with* $f(c) = y$*.*

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Let $J = f([a, b])$.
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Let y be between $f(a)$ and $f(b)$, and suppose that $y \notin J$.
Without loss of generality, assume $f(a) < f(b)$.

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Let y be between $f(a)$ and $f(b)$, and suppose that $y \notin J$.
Without loss of generality, assume $f(a) < f(b)$.
Then $J = U \cup V$ for

$$U = (-\infty, y) \cap J \quad \text{and} \quad V = (y, \infty) \cap J$$

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These are nonempty, disjoint, and open in the subspace J of $\mathbb{R}$.

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This is a contradiction! □

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Uniform continuity

Definition

Let (S, d_S) and (T, d_T) be metric spaces and $A \subseteq S$. We say that a function $f : A \to T$ is **uniformly continuous** on A if for all $\epsilon > 0$ there exists $\delta > 0$ such that

$$d_S(x, y) < \delta \Rightarrow d_T(f(x), f(y)) < \epsilon$$

for all $x, y \in A$.

- f continuous on A means f is continuous at every $a \in A$
- δ can depend on ϵ and a (not on x)
- with uniform continuity, δ can't depend on a ...

Examples

- x is uniformly continuous on $\mathbb{R}$
- x^2 is NOT uniformly continuous on $\mathbb{R}$
- $1/x$ is continuous on $(0, 1)$
- $1/x$ is NOT uniformly continuous on $(0, 1)$

Relationship with compactness

Theorem (Heine)

Let (S, d_S) and (T, d_T) be metric spaces and $A \subseteq S$ be compact. Suppose that $f : A \to T$ is continuous on A, then f is uniformly continuous on A.